Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	01 October 2026
Team ID	SWTID-2026-8643
Project Name	Script-Controlled ACL – Control Record Access Based on Field Value (ServiceNow)
Maximum Marks	4 Marks

Functional Requirements:

The proposed solution has the following functional requirements.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User & Role Management	Create the test user (EEE User) Create custom roles bb1, bb2, bb3, bb4 Assign all roles to the EEE User
FR-2	Data Management	Create custom table Institution Details (u_institution_details) Add fields: Student Roll Number (auto number), Student Name, Faculty Name, Branch (ECE, EEE, CSE), Email, Phone Number, Description Create sample records for different branches
FR-3	Read Access Control	Read ACL requiring role bb1 with data condition Branch is EEE and an ACL script (gs.hasRole) Admin can view all records Users without the role see no records
FR-4	Create, Write and Delete Access Control	Create ACL requiring bb2 (New button) Write ACL requiring bb3 (edit records) Delete ACL requiring bb4 (delete with confirmation)
FR-5	Verification	Elevate security_admin to create ACLs Impersonate EEE User, a user without roles and Admin Confirm the list shows only the records each user may access

Non-functional Requirements:

The proposed solution has the following non-functional requirements.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Uses the standard ServiceNow list and form interface; buttons such as New, edit and Delete appear only when permitted; users without access see a clear "No records to display" message.
NFR-2	Security	Record-level READ, CREATE, WRITE and DELETE access controls with role-based access (bb1–bb4); script-controlled decision that denies access by default; security_admin elevation required to modify ACLs; admin override retained.
NFR-3	Reliability	ACLs are enforced consistently on every list, form and portal request, so access results are the same each time for the same user and record.

NFR No.	Non-Functional Requirement	Description
NFR-4	Performance	ACLs are evaluated on the server using simple role and data conditions and a short script, so lists load quickly without noticeable delay.
NFR-5	Availability	The solution runs on a cloud-hosted ServiceNow instance whose hosting and availability are managed by the platform.
NFR-6	Scalability	The same ACL pattern can be reused for more branches, roles or tables by adding roles and conditions, without rewriting the application.